6.4.1 Irreducibility of Finite Markov Chains

The concept of ergodicity, and the method of analyzing it, can be understood using an analogy to a simpler finite model, a discrete Markov Chain.

A discrete Markov Chain is a stochastic process in which, at each step, a transition is made from the current state i to another j selected from a given probability distribution. In the case of a finite collection of states $\Sigma = \{1, 2, \ldots, k\}$, we assume that the transition probabilities $\pi_{ij} \in [0, 1]$ are given and denote the state at time n by S_n. Thus we have

$$\Pr(\mathrm{S}_{n+1} = j | \mathrm{S}_n = i) = \pi_{ij},$$

where we denote by $\boldsymbol{\Pi} = (\pi_{ij}) \in \mathbb{R}^{k\times k}$ the matrix of transition probabilities, i.e. it is a matrix whose entries are all nonnegative and less than or equal to 1. The sum of the elements in row i gives the probability of being in *some* state at time $n + 1$ given that we were in state i at time n, thus it is one. The matrix $\boldsymbol{\Pi}$ is sometimes referred to as a *right stochastic matrix*.

A distribution on the state space is typically associated to a row vector $\boldsymbol{\psi} = (\psi_1, \psi_2, \ldots, \psi_k)$ with nonzero elements and which is normalized such that

$$\sum_{i=1}^{k} \psi_i = 1.$$

This might be viewed as defining the relative proportion, in a large collection of *walkers* (independent realizations of a stochastic evolution), that are found in each state. As the walkers make moves based on the transition probabilities, the distribution will vary. We may denote by $\boldsymbol{\psi}_n = (\psi_{n,1}, \psi_{n,2}, \ldots, \psi_{n,k})$ the distribution observed at step n of the iteration. The probability of being in state i at the $n + 1$st iteration, is the sum over j of the probabilities of being in state j multiplied by the probability of transitioning from state j to state i, thus

$$\boldsymbol{\psi}_{n+1} = \boldsymbol{\psi}_n \boldsymbol{\Pi}.$$

This iteration can be viewed as analogous to dynamics. We may think of the evolution of the Markov chain as being realized through multiplication by the Markov matrix $\boldsymbol{\Pi}$. Let $\pi_{ij}^{(l)}$ represent the transition probability of being in state i at iteration l given that we were in state j at time 0, i.e. $\pi_{ij}^{(l)}$ is the (i, j) entry of the matrix $\boldsymbol{\Pi}^l$.

Two important properties of Markov chains are aperiodicity and communication. For each state of the chain, we define the number $d(i)$ to be the greatest common divisor of all integers l such that $\pi_{ii}^{(l)} > 0$ (if the probability of return to state i is zero, then we set $d(i) = 0$). If $d(i) = 1$, for all i then we say that the chain is *aperiodic*. States i and j are said to *communicate* if $\pi_{ij}^{(l_1)} > 0$ and $\pi_{ji}^{(l_2)} > 0$ for

$$\begin{bmatrix} 0.2 & 0.1 & 0.7 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0.3 & 0 & 0 & 0.3 & 0.4 \\ 0 & 0.5 & 0 & 0.3 & 0.2 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

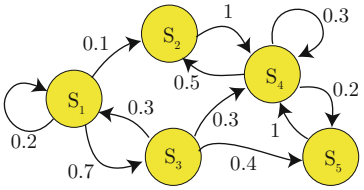


Fig. 6.6 A Markov matrix (*left*) and its graph representation (*right*). In this example, the Markov Chain is not irreducible

some integers l_1 and l_2. This means that it is possible to reach j from i with positive probability and vice-versa.

A finite state space model is *irreducible* if every state is reachable from every other state in a finite "time", where time in this context refers to the number of iterations of the chain. Equivalently, we say the model is irreducible if all states communicate.

A very useful representation of the Markov Chain transition matrix $\boldsymbol{\Pi}$ is by use of a weighted directed graph in which each node represents a state of the Markov Chain and the weights represent transition probabilities. Then one may understand irreducibility as expressing the connectedness[8] of this graph (see Fig. 6.6).

An *invariant distribution* is a left eigenvector of $\boldsymbol{\Pi}$, having only non-negative elements, and corresponding to the eigenvalue 1. That is, an invariant distribution $\boldsymbol{\psi}_*$ satisfies

$$\boldsymbol{\psi}_* = \boldsymbol{\psi}_* \boldsymbol{\Pi}\,.$$

The fundamental result for Markov chains is the following: *If a given Markov chain is both irreducible and aperiodic, then it can be shown that there is a unique invariant distribution* $\boldsymbol{\psi}_*$ *such that*

$$\lim_{n\to\infty} \boldsymbol{\psi}_n = \boldsymbol{\psi}_*.$$

for arbitrary initial distribution $\boldsymbol{\psi}_0$. The convergence of an irreducible finite Markov chain to its stationary distribution can be understood through the properties of its eigenvalues by applying of the Perron-Frobenius theorem (see [269]).

[8] We say that a graph is *connected* if every node is linked to every other node by a sequence of contiguous edges. For a directed graph, we must have, for any pair of vertices A and B, paths which take us from A to B and vice-versa.